Agentic Crystallization v1.5

Cost-Aware Information Acquisition: A Falsifiable Adaptive-Procedure Study

Status: Reproducible synthetic research. No phase transition or autonomous Pegasus self-improvement established.

Abstract

Versions 1.0–1.4 examined planted higher-order procedure interactions, misleading connectivity, conditional transfer, negative transfer, and Bayesian updates. Version 1.5 introduces a one-step value-of-information (VOI) controller deciding whether a paid relevance cue is worth requesting. Across 720 new held-out synthetic worlds, the controller improves mean utility relative to static Bayes when 75% of tasks are related (+1.597), but loses when 25% are related (–0.699). This isolates a genuine limitation of using an imperfect prior to value additional computation.

1. Hypothesis ledger

- H1: Nonadditive procedure value can be measured; supported on planted synthetic cases only.

- H2: A nonlinear critical adaptive transition; not demonstrated.

- H3: Validated clusters improve later learning; only restricted synthetic transfer has been observed.

- Product question: Can a bounded controller avoid unproductive tool calls? Tested here in an oracle simulation only.

2. Mathematical continuity

For procedure utility J(S;D), the higher-order finite-difference interaction is

```
\Gamma_T(S)=\sum_{A\subseteq T}(-1)^{|T|-|A|}J(S\cup A;D).
```

A three-way AND task has a nonzero triple interaction at the empty baseline but zero empty-baseline pairwise terms. This is standard Möbius algebra, not a novel identity.

The proposed engineering potential

```
F(S)=\alpha B(S)+\kappa V(S)-\beta U(S)
```

relates measured coordination, verification and future utility. It is not thermodynamic free energy. Graph connectivity and adaptation efficiency remain separate.

3. New exact one-step VOI derivation

Let R be binary task-family relatedness, q=P(R=1|h), and L_r(t) the conditional probability of remaining candidate t under R=r. The candidate probability is

```
p(t|h)=(1-q)L_0(t)+qL_1(t).
```

For a candidate order \(\pi=(t_1,\ldots,t_k)\), a remaining query budget b, success reward W=40 and query cost c=0.1, expected continuation utility is

```
V_b(\pi,p)=W\sum_{i=1}^{k}p(t_i)-c\left[\sum_{i=1}^{k}i\,p(t_i)+k\left(1-\sum_{i=1}^{k}p(t_i)\
right)\right],
\quad k=\min(b,|\pi|).
```

This accounts for failed and censored searches. Sorting candidates by descending p is optimal under this equal-cost exact-match model.

A binary cue C with accuracy a updates q through Bayes' rule:

```
q_c=\frac{qP(C=c|R=1)}{qP(C=c|R=1)+(1-q)P(C=c|R=0)}.
```

The model-estimated net value of probing is

```
\operatorname{VOI}(h)=\sum_c P(C=c|h)\max_\pi V_{b-1}(\pi,p_c)-c-\max_\pi V_b(\pi,p).
```

Probe only when VOI>0, before observing C. The cue consumes one of the same 40 attempts. The controller does not see the hidden target or latent relatedness.

Under a fixed correct joint target distribution, eliminating wrong exact-match candidates only renormalizes survivors; it cannot reorder their relative probabilities. A new cue can reorder them.

4. Experiment contract

- Training: frozen Stage F conditional likelihood, seeds 3000–3399; no Stage G holdout updates.

- Holdout: seeds 8000–8119; three catalog sizes 12,15,18; prevalence 0.25 and 0.75; 720 synthetic worlds.

- Oracle: exact triple match. First validated triple is supplied, with discovery cost charged equally across policies.

- Cue: independent synthetic 80%-accurate relevance signal, only revealed if paid for.

- Policies: no prior, always reuse, static Bayesian, always probe after eight misses, and VOI.

- Budget: 40 attempts including any paid cue.

- Utility: 40*found – 0.1*(first discovery cost + second attempts).

- Data: world-paired rows, fixed seeds, manifests, and SHA-256 receipts.

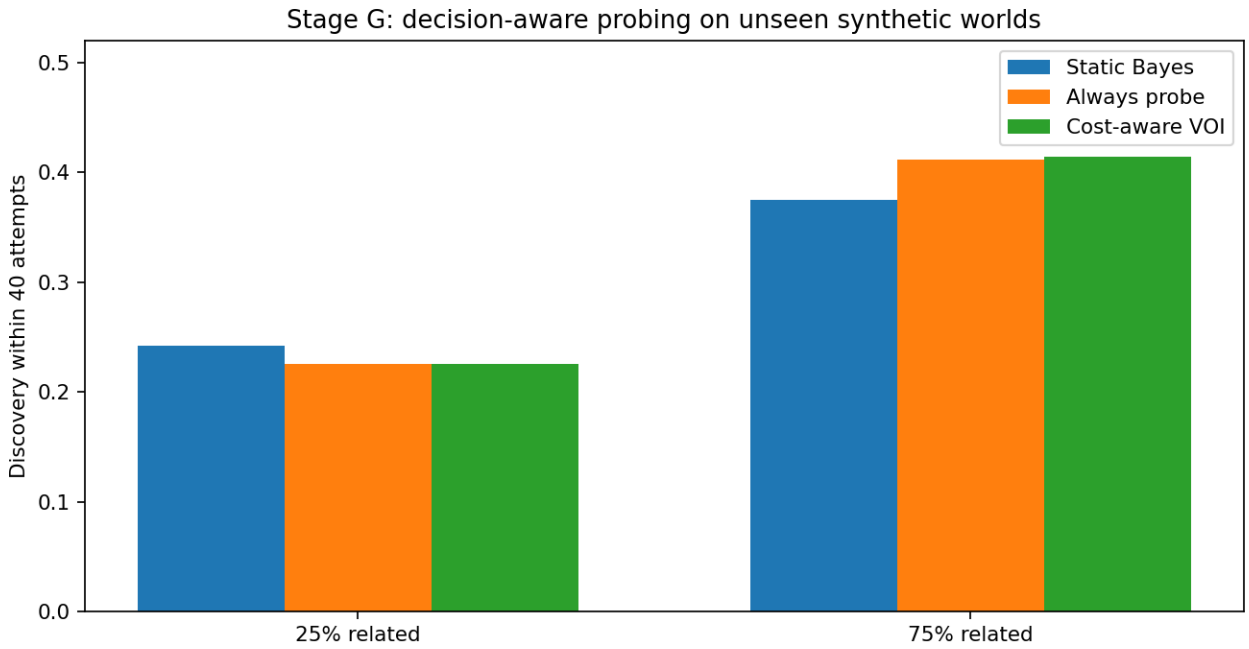
5. Results

Related-task prevalence	Static success	Always-probe success	VOI success	VOI probe rate	VOI minus static utility	VOI minus always-probe
25%	24.17%	22.50%	22.50%	70.56%	**–0.699**	+0.005
75%	37.50%	41.11%	41.39%	48.06%	**+1.597**	+0.127

Descriptive paired bootstrap intervals across worlds (3,000 resamples, not confirmatory):

- 25%: VOI – static –0.699 [–1.753,+0.330]; VOI – always probe +0.005 [+0.003,+0.008].

- 75%: VOI – static +1.597 [+0.790,+2.514]; VOI – always probe +0.127 [+0.013,+0.354].



6. Interpretation and falsification

Positive: In the high-relatedness regime, an explicitly priced information decision improves mean utility over both the static and fixed-probe policies.

Negative: In the low-relatedness regime, VOI probes too often relative to static Bayesian search. A model-positive VOI does not guarantee a positive realized value under prior mismatch. This falsifies any claim of universal advantage for the proposed controller.

Important: The likelihood model uses a pooled relatedness prior trained on different mixtures; it does not know the true test prevalence. These results do not demonstrate phase transitions, autonomous procedure discovery, learning-to-learn, or performance gains in Pegasus.

7. Next experimental step

Test calibrated, distributionally robust VOI: estimate task-relatedness prevalence from training-visible context, include cue reliability uncertainty, require a conservative positive margin, and compare with strong static/adaptive controls on untouched gradual-overlap and shifted distributions. Track regret, cost, failed attempts, and abstentions. Study H2 independently with graph/hypergraph finite-size sweeps.

8. Reproduction

```
PYTHONPATH=src OPENBLAS_NUM_THREADS=1 python -m unittest discover -s tests -v
PYTHONPATH=src python -m agentic_crystallization.stage_g --out experiments/reproduced_stage_g -
-start-seed 8000 --count 120
```

On Windows PowerShell: $env:PYTHONPATH='src' then run the two python -m commands.

[Source](../src/agentic_crystallization/stage_g.py) · [Tests](../tests/test_stage_g.py) · [Raw data](../experiments/stage_g_holdout/runs.csv) · [Manifest](../experiments/stage_g_holdout/manifest.json) · [Prior paper](archive/WHITE_PAPER_1.4.md).